Papulosquamous Disorders

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1 Introduction

The skin diseases listed here are initially reaction patterns of the skin characterized by inflammatory vasodilatation (erythema), to which a cellular infiltration, usually by T cells and neutrophilic granulocytes, and epidermal changes can be added. These lead clinically to papules or plaques and to some characteristic scaling (e.g., psoriasiform or pityriasiform). The classification is not strictly systematic, and it is not clear in all cases whether they are independent clinicopathological disease entities. An example is erythema annulare centrifugum. Other diseases that can be assigned to this group are dealt with in other chapters (e.g., ▶ Chap. 39, "Psoriasis," or ▶ Chap. 40, "Papular and Lichenoid Diseases"). Inflammatory skin disease in which the immigration of inflammatory cells leads to macroscopically visible pustules, such as erythema toxicum neonatorum, are also excluded here.

The classification of the inflammatory diseases discussed in this chapter is based on the clinical description. A future path to a modern classification could take advantage of other dimensions of the inflammatory process besides or even beyond the clinical picture of typical lesions (Fig. 1). These include:

- The histological pattern of inflammation, for example, eczematous, psoriasiform, lichenoid, granulomatous (Table 1),
- The essential cells of the infiltrate, for example, T cells, neutrophilic granulocytes, eosinophilic granulocytes, histiocytes (Table 2), and
- At the subcellular level, the activated signal transduction pathways or essential molecules of signal transduction, for example, cAMP, Janus kinases (JAK1-3, TYK2), signal transducer and activator of transcription proteins (STAT1 to STAT6), and finally
- The dominant cytokine pathways, for example, tumor necrosis factor-α, interferon-γ, interleukin-4, -13, -17, -22, and -23.

Several of the inflammatory reactions described here are probably based on exaggerated cutaneous defense programs in which epidermal repair mechanisms are also activated. This results in characteristic changes of the epidermal microanatomy, which are then part of the histological pattern. Examples are extensive hyperparakeratosis and acanthosis with hypogranulosis in psoriasis, in contrast to orthohyperkeratosis and irregular, often sawtooth acanthosis with hypergranulosis in lichen planus.

In the case of psoriasis (▶ Chap. 39, "Psoriasis"), the identification of these different levels of inflammatory reaction has already been quite successful. In the case of the skin inflammations described here, a typical histological basic pattern and occasionally also a dominant cytokine pattern is also present in addition to the clinical picture. Another intention of this approach is that it will be easier to find suitable therapeutic strategies, as there are no high-quality clinical studies available for most of the diseases described here, and no drugs approved for their treatment.

Little is known about the specific etiology of the skin lesions discussed in this chapter. There are some overlaps with regard to the triggering factors (for example, the triggering factors for erythema annulare centrifugum, erythema multiforme, and Sweet's syndrome), without yet knowing under what circumstances the respective skin reactions develop in different patients.

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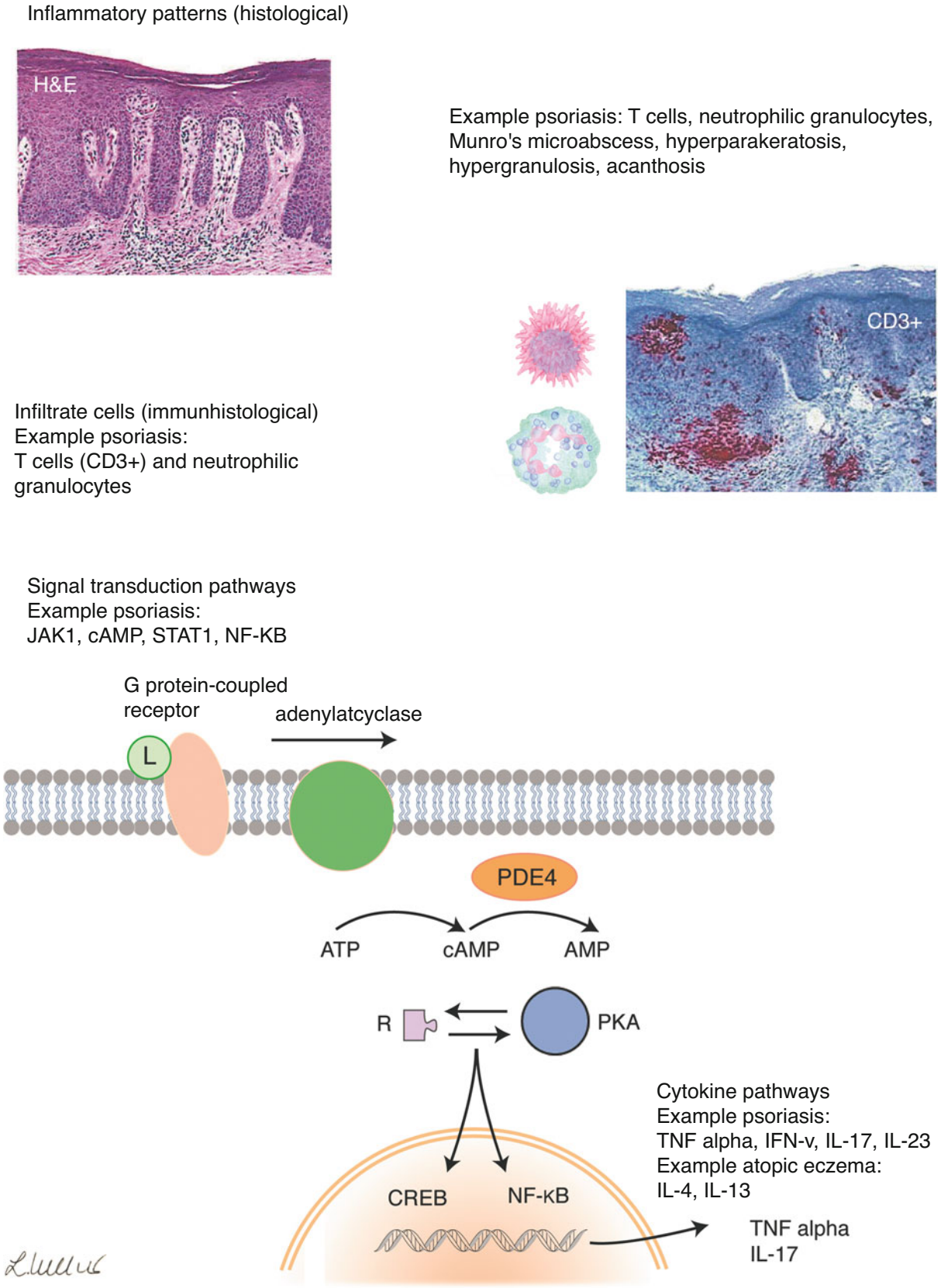


Fig. 1 Concept of the dimensions of cutaneous inflammation: histological pattern (inflammatory infiltrate, epidermal changes), infiltrating cells, dominant signal transduction pathways, and cytokine patterns (▶ Chap. 39, "Psoriasis" for psoriasis, ▶ Chap. 40, "Papular and Lichenoid Diseases" for lichen planus, and ▶ Chap. 32, "Atopic Dermatitis")

Table 1 Selection of histological cutaneous inflammatory patterns

Inflammatory patterns (histological)	Occurrence in (examples)
Psoriatic inflammation	Psoriasis, pityriasis rubra pilaris, Reiter's disease, chronic eczema
Lichenoid inflammation	Lichen planus, pityriasis lichenoides, collagenoses, drug reactions
Eczematous inflammation	Atopic eczema, contact allergy, drug reaction
Granulomatous inflammation	Intracellular pathogens, foreign body reaction, sarcoidosis, Crohn's disease, rheumatic nodules

Table 2 Important cell types involved in skin inflammation

Cell type(s)	Classification and function
Dendritic cells	Classification epidermal, dermal, functional: TIP-DC (TNF-α and iNOS producing DC), IDEC (inflammatory dendritic epidermal cell), DC1 (activated Th1 cells), DC2 (activated Th2 cells); activation of T cells, B cells, immune balance
Macrophages, histiocytes	Mobile and tissue-resident phagocytes and antigen-presenting cells
T cells	Possible classification according to the T cell receptor αβ or γδ cells, the expression of surface markers, for example, $CD4^+$ helper cells, $CD8^+$ cytotoxic cells or the expression of essential transcription factors and cytokines, for example, Th1 (T-bet; IFN-γ), Th2 (GATA3; IL-4), Th17 (RORγt; IL-17), Treg cells (Foxp3; TGFβ)
Innate lymphoid cells (ILC)	Classification according to cytokines produced; group 1 (IFNγ, TNF), group 2 (IL-4, IL-5, IL-9; IL-13), group 3 (IL-17A, IL-22)
B cells	Limited knowledge about the specific role in cutaneous inflammation; antigen presentation and antibody production
Granulocytes	Neutrophilic or eosinophilic granulocytes
Mast cells	Local release of multiple immediate-type mediators and cytokines, neuroinflammation
Keratinocytes	Epidermal component: Ki-67 indicates proliferating cells, cytokeratin-16 abnormal differentiation; but also nonprofessional immune functions, for example, production of chemoattractive cytokines such as IL-8 (CXCL8)

For some representatives of the group, however, a classification other than the purely clinical one is possible, such as the classification of necrolytic migratory erythema and the flushing in carcinoid syndrome as cutaneous paraneoplasia (▶ Chap. 114, "Paraneoplastic Disorders").

2 Erythemas

The reddening of the skin due to vasodilatation is one of the main skin changes assigned to this group; infiltration or scaling is variable. Some representatives of this group, which are exclusively characterized by an erythema, are discussed in ▶ Chap. 63, "Functional Angiopathies." Examples are the persistent facial erythema and the familial or acquired palmoplantar erythema.

2.1 Figurate Erythemas

This group of erythematous skin diseases is characterized by the occurrence of figurate or annular red lesions with a tendency toward centrifugal spread. Although some have been described as independent entities, newer concepts are more concerned with reaction patterns that can occur mainly in allergic reactions, collagenoses, and paraneoplasms. Diseases in which figurate erythema is observed include:

- Dermatophyte infections
- Urticarial reactions, including early urticarial stage of bullous pemphigoid
- Erythema multiforme
- Annular (pustular) psoriasis
- Lupus erythematosus, especially subacute cutaneous and neonatal lupus erythematosus
- Sjögren syndrome, especially in Asian patients
- Lymphomas

2.1.1 Erythema Annulare Centrifugum

(Darier 1916)

Epidemiology

Erythema annulare centrifugum occurs mainly in middle-aged adults without sex preference.

Etiopathogenesis

It has a clinically defined reaction pattern rather than being an independent entity, even if other diseases cannot always be detected. In a retrospective analysis of 90 cases, 22% could be assigned to another diagnosis (urticaria, leukocytoclastic vasculitis, psoriasis), 32% to lupus erythematosus, and 18% to pseudolymphoma, of which the majority of patients had Lyme disease. The following overview shows diseases associated with erythema annulare centrifugum.

Diseases Associated with Erythema Annulare Centrifugum

- **Malignant tumors:** B cell lymphomas, chronic lymphatic leukemia, mycosis fungoides, but also many other malignancies; remission was observed after excision of malignant tumors (e.g., breast, gastrointestinal tract, lungs) and recurrence of skin symptoms after relapse.
- **Infections and parasitic diseases:** herpes zoster, intestinal candidosis, syphilis, borreliosis, HIV, mycid and differential diagnosis in tinea corporis; other bacterial infections, as well as infestation with ascarids or other worms, were also associated with the occurrence of skin signs.
- **Drug reactions:** hypersensitivity reactions with the appearance of an erythema annulare centrifugum have been described with various drugs, including: diuretics (hydrochlorothiazide, aldactone), nonsteroidal anti-inflammatory drugs (salicylates, piroxicam), chloroquine, cimetidine, amitriptyline, etizolam, finasteride.
- **Food allergies:** reactions to fish, peanuts, and cheese (especially blue or other types of mold cheese) can occur with the appearance of an erythema annulare centrifugum.
- As a manifestation of an **autoimmune progesterone dermatitis** and during p**regnancy** (third trimester).
- **Chronic inflammatory/rheumatic diseases:** lupus erythematosus, sarcoidosis, relapsing polychondritis, appendicitis.

In the majority of the cases described, a temporal but not a definite causal connection can be observed.

Clinical Features

On the trunk, also on the proximal extremities and the gluteal area, but more rarely on the face, on the head, or at acral locations mostly asymmetrically elevated, partly urticarial erythema develops, which slowly expands centrifugally while central clearing occurs (Fig. 2). In this way, arcuate, circular or polycyclic limited band-shaped lesions gradually emerge, some of which can lose their original character within hours and then only appear erythematous. The changes can recede after days, weeks or months. New lesions develop in areas not yet affected but also in areas that have cleared. The centrally cleared areas can be secondarily slightly hyperpigmented. As a rule, an inwardly directed (collarette-like) scaly ruffle is visible in the marginal area; fine blisters or hemorrhages are also rarely visible. Patients often have few symptoms; particularly in urticarial elevated erythema, itching may be present. The occasional patient may have fever accompanying new lesions.



Fig. 2 Erythema annulare centrifugum

Differential Diagnosis

According to today's understanding, the first question should be whether an erythema annulare centrifugum or an annular variant of another disease is actually present, such as urticaria, tinea corporis, erythema multiforme, lupus erythematosus, Sjögren's syndrome, psoriasis, dermatitis herpetiformis, bullous pemphigoid, or pemphigus. It should also be taken into account that the majority of suspected cases actually turn out to be an annular lupus erythematosus tumidus (▶ Chap. 54, "Lupus Erythematosus") or a B cell pseudolymphoma (▶ Chap. 109, "Pseudolymphomas"). Finally, other diseases with annular manifestation must be excluded, including granulomatous diseases such as granuloma annulare or annular forms of sarcoidosis (lupoid infiltrate, diascopy) and leprosy. In erythema gyratum repens (Gammel's syndrome), erythema migrates rapidly. Erythema papulatum centrifugum is similar, but more papular and has so far only been described in Japan. In childhood, other forms of annular erythema can be considered as differential diagnosis.

Histopathology

The hallmark is a prominent lymphocytic perivascular infiltrate in the dermis. If the infiltrate is relatively superficial, minimal epidermal changes (spongiosis, scale) may be found.

Eosinophils may be present. The former division into a superficial and a deep form is considered obsolete. The deep form is identical with lupus erythematosus tumidus or pseudolymphoma.

Therapy
It is important to treat the specific skin disease and if necessary the underlying disease. The treatment depends on severity and persistence. Topical or systemic glucocorticoids or vitamin D analogs in combination with phototherapy (e.g., UVB 311 nm) were described as therapy of erythema annulare centrifugum.

2.1.2 Erythema Gyratum Perstans
(Colcott Fox 1885)

Synonym
Familial annular erythema (Beare et al. 1966)

This very rare disease is inherited in an autosomal-dominant pattern (OMIM 106500). The lesions may begin shortly after birth or first appear in childhood.

It is a chronic disease usually without further systemic involvement. An association with familial neurological disorders is described in a case report. The skin changes are clinically similar to erythema annulare centrifugum.

2.1.3 Annular Erythema of Infancy
In addition to erythema gyratum perstans, there are a number of other figurate erythemas described in infancy. Annular erythema of infancy typically features small papules that evolve into erythematous arches or rings without scales. Lesions last for several days and recur in cycles. Microscopic examination shows a perivascular lymphocytic infiltrate often with eosinophils. This disease is probably closest to erythema annulare centrifugum.

In some cases, central atrophy develops; this is known as **erythema gyratum repens atrophicans transiens** (Gianotti and Ermacora 1975). Other authors see this skin change as a variant of neonatal lupus erythematosus. In other cases of annular erythema the histological picture shows similarities to leukocytoclastic vasculitis with immigration and decay of neutrophils. For this special form, the term **neutrophilic figurate erythema of infancy** (Annessi et al. 1997) has been suggested. The closest adult equivalent appears to be urticarial vasculitis. Usually, it shows a benign self-limiting course over months without systemic involvement. In addition, erythema annulare rheumaticum is mostly observed in infants (see below). Finally, one must rule out neonatal lupus erythematosus (▶ Chap. 54, "Lupus Erythematosus"), which is often annular. A fungal infection by dermatophytes or *Malassezia* species should be considered.

In all cases except when *Malassezia furfur* is found, watching and waiting rather than any specific treatment is the only recommended action.

2.1.4 Erythema Annulare Rheumaticum
(Lehndorff and Leiner 1922)

Synonyms
Erythema marginatum (rheumaticum); Erythema circinatum

Epidemiology
About 10–20% of patients with acute rheumatic fever develop a transient erythematous rash. Most patients are children with cardiac involvement, which can also occur without joint involvement. Similar skin changes can also occur in other diseases, such as hereditary angioedema.

Etiopathogenesis
Erythema annulare rheumaticum is probably a hypersensitivity reaction in genetically predisposed individuals. After infection with hemolytic group A streptococci, a Th1-cell-dominated inflammation of various organs (joints, kidney, heart, central nervous system, skin) can occur because of a cross-reaction with the body's own antigens.

Clinical Features
The most common site of involvement is the trunk. The initial lesions are pink macules that rapidly, over a period of minutes, spread to produce a polycyclic or serpiginous pattern (Fig. 3). The bands of erythema are typically quite narrow; thus, the final pattern has been compared to chicken wire. Typically, the cutaneous lesions appear along with a fever spike, often in the late afternoon. They usually disappear within hours to days, are asymptomatic, and easily overlooked. Erythema marginatum is more common in patients with active carditis. It usually develops at about the



Fig. 3 Erythema annulare rheumaticum

same time as the typical joint and cardiac findings, but may appear weeks later.

Two other cutaneous manifestations of rheumatic fever are:

- **Erythema papulatum (**Cockayne 1912). Tiny erythematous urticarial macules and papules appear on the knees and elbows. They do not evolve into annular lesions and resolve over a period of days to weeks. They clinically resemble granuloma annulare and are present in less than 5% of patients.
- **Subcutaneous nodules**. Patients with more severe carditis may develop nodules over the extensor tendons of the extremities, the scalp, and the spine. Histologically, these nodules show extensive necrobiosis and are identical to granuloma annulare or rheumatoid nodules, but they tend to disappear in a matter of months (▶ Chap. 42, "Granulomatous Diseases").

Differential Diagnosis

Patients with juvenile rheumatoid arthritis (Still's disease) may also have a transient rash, similar to erythema marginatum. More often they have erythematous macules, usually on the extremities, that are relatively stable and not annular.

Histopathology

There is a sparse perivascular infiltrate of neutrophils in the papillary dermis. This is a relatively uncommon finding that may help in a puzzling case.

Therapy

The underlying rheumatic fever should be treated with antibiotics. The skin findings do not require any special therapy.

2.1.5 Erythema Scarlatiniforme Desquamativum Recidivans

(Féréol 1876; Besnier 1890)

Etiopathogenesis

According to today's understanding, this very rare disease can be assigned to staphylococcal or streptococcal toxin-mediated erythema. Drug reactions have also been described as triggers (▶ Chap. 35, "Cutaneous Adverse Drug Reactions"). The pathophysiological principle can be explained by recurrent toxin-mediated perineal erythema (RTPE) (▶ Chap. 12, "Staphylococcal and Streptococcal Diseases of the Skin"). In the area of the perineum, groin, and lower abdomen, a sharply defined scaly rash appears. Axillary erythema, acral edema, or mucositis and raspberry tongue may be present. The rash is asymptomatic or associated with mild itching or burning. Swabs of the affected skin areas are negative. Throat swabs usually reveal superantigen-producing staphylococci (e.g., TSST-1) or group A streptococci (streptococcal pyrogenic exotoxin A or B).

Clinical Features

The prodrome includes fever, chills, and headache. This is followed by a macular exanthem that starts usually on the trunk and over days becomes generalized. The hands, feet, and head are the last to be involved. As the erythema fades, there is characteristic desquamation, with the typical sock and glove shedding on the palms and soles, as well as larger scales on the trunk. In the localized form, only the hands and feet are affected: erythema scarlatiniforme desquamativum recidivans localisatum. Skin findings usually resolve over a period of weeks, but may reoccur.

Differential Diagnosis

All scarlatiniform eruptions should be considered including scarlet fever as well as tinea manuum and pedum, dyshidrotic eczema, pityriasis rubra pilaris, and psoriasis.

Therapy

If an infectious disease is present, it should be treated with antibiotics. Otherwise, topical anti-inflammatory and keratolytic agents can be used.

2.2 Paraneoplastic Erythema

2.2.1 Erythema Gyratum Repens

The paraneoplastic skin disease erythema gyratum repens usually occurs in malignancies of inner organs. It is characterized by rapidly migrating, annular polycyclic lesions, which may look like wood grain. Erythema gyratum repens-like skin lesions have also been described in some cases in other diseases, the spectrum being comparable with the associations observed for erythema annulare centrifugum; for example, psoriasis, pityriasis rubra pilaris, urticaria vasculitis, autoimmune blistering dermatoses, lupus erythematosus, and cutaneous lymphomas (▶ Chap. 114, "Paraneoplastic Disorders").

2.2.2 Necrolytic Migratory Erythema

Synonyms

Erythema necroticans migrans; Glucagonoma syndrome

This rare eruption is almost an obligate paraneoplasia and almost all patients have glucagonoma. Most are α-cell tumors; 80% are malignant but usually slowly growing. Very rarely, patients have been described without a malignancy, some with underlying gastrointestinal disease (chronic pancreatitis, chronic hepatitis, carcinoma of the colon) and others in apparently good health (pseudo-glucagonoma syndrome). Although both glucagon itself and the resulting hypoaminoacidemia have been blamed for the eruption, the

exact cause remains unclear (▶ Chap. 114, "Paraneoplastic Disorders").

2.2.3 Flushing in Carcinoid Syndrome

Flushing is triggered by the production of vasoactive substances such as serotonin or bradykinin in neuroendocrine tumors (e.g., gastrointestinal carcinoids; ▶ Chap. 99, "Endocrine Diseases").

2.3 Erythema Multiforme

(Hebra 1866)

Synonym

Erythema exsudativum multiforme (EEM)

Definition

Erythema multiforme is an acute mucocutaneous reaction with many possible triggering agents, but characteristic symmetrical acral target or iris lesions. Two forms are identified, defined by severity, clinical course, and trigger: erythema multiforme minus (EM minus), limited to cutaneous involvement and often triggered by infections, and erythema multiforme majus (EM majus) with mucosal lesions and a cytotoxic hypersensitivity reaction to drugs. It is therefore described elsewhere together with Stevens–Johnson syndrome and toxic epidermal necrolysis (▶ Chap. 36, "Severe Cutaneous Adverse Drug Reactions").

Erythema multiforme affects primarily young adults. Severe EM majus is most common in children and adolescents. It may occur more frequently after mycoplasma infections, in half of the cases with mucosal involvement.

Epidemiology

Erythema multiforme usually occurs between the second and fourth decades of life. Men are more often affected than women.

Etiopathogenesis

Erythema multiforme is triggered by infections and vaccinations. Although the mechanisms are not clear, a delayed-type hypersensitivity reaction is most likely. Severe drug reactions such as Stevens–Johnson syndrome may start as EEM. The most common causes are:

- **Viral infections:** the most common cause of EM minus is herpes simplex virus infections. Almost all recurrent EM minus is herpes-related. HSV type I is far more frequently associated with EM than HSV type II. Occasionally, HSV-DNA can be identified in erythema multiforme lesions. Other viruses, such as parvovirus B19, VZV, HCV, EBV, Orf virus, and SARS-CoV2, may cause usually nonrecurrent erythema multiforme.
- **Vaccinations:** occasionally, an EM was observed after vaccination with the polyvalent measles mumps rubella vaccine, after hepatitis B vaccination and vaccination with polyvalent HPV vaccine.
- **Other infections:** the most common cause of EM majus is respiratory infections, primarily *Mycoplasma pneumoniae*. Rarely, other airway infections may lead to milder erythema multiforme; EM has been described in syphilis, especially in patients with simultaneous HIV infection, and in association with infections by *Trichophyton mentagrophytes*, *Histoplasma capsulatum*, or *Gardnerella vaginalis* and *with* vulvovaginal candidiasis
- **Allergic contact dermatitis:** allergic contact EM has been described (paraphenylenediamine in tattoos, medicinal herbs such as *Alpinia galanga*, topical glucocorticoids).
- **Chronic inflammatory diseases, autoimmune diseases:** Kawasaki syndrome, Crohn's disease, lupus erythematosus (Rowell's syndrome), graft-versus-host-disease (GVHD)
- **Hormones:** EM rarely occurs parallel to menstruation; cases of progesterone-dependent EM have been described.
- **UV-provoked**: EM-like polymorphic light eruption, EM-like photoallergic reaction after medication
- **Malignancies:** for example, T cell leukemia, cholangiocarcinoma

In up to 60% of cases no trigger is identified. A concept explaining the connection between various triggers, immunological reaction patterns, and the characteristic clinical picture is still lacking.

Clinical Features

Erythema multiforme minus or classic erythema multiforme has symmetric, target- or iris-shaped lesions, almost invariably involving the hands, especially the dorsal aspects. More severe disease may affect the arms, feet, neck, and lateral aspects of the face. Children and adolescents may develop lesions on the trunk.

The typical lesion is initially an erythematous macule that enlarges and becomes annular. Within 1–2 days, the lesions are about 1 cm in size, with a central area of hemorrhage, necrosis, and blistering, surrounded by a pale cyanotic region and then enclosed by an erythematous peripheral band. This combination produces the classic, disease-defining target or iris lesion. Lesions may coalesce, evolve into polycyclic



Fig. 4 Erythema multiforme

patterns, and show central healing. Other lesions may lack a target pattern and be more urticarial, whereas in others the blisters dominate (Fig. 4).

Erythema multiforme majus is characterized by similar skin lesions accompanied by subepidermal blisters and hemorrhagic erosions involving the lips, mouth, conjunctiva, and genital mucosa (▶ Chap. 34, "Pruritus and Prurigo"). The mucosal lesions can precede, accompany, or follow the cutaneous changes. More severe erosions may heal with hyperpigmentation. When the mucosal erosions precede cutaneous target lesions, the diagnosis is very difficult. When acute herpes simplex involves the lips, it can be impossible to separate late herpes simplex erosions from early EM majus erosions. The mucosal lesions also usually heal without scarring, but extensive oral involvement can lead to adhesions in the corner of the mouth. Similarly, genital involvement may cause phimosis or strictures. Ocular involvement is especially dangerous, with irreversible damage that cannot be distinguished from Stevens–Johnson syndrome or toxic epidermal necrolysis.

Differential Diagnosis

Once target lesions are present, the diagnosis is easy. When there is truncal involvement or less typical lesions then urticaria, early bullous pemphigoid, Sweet's syndrome, or a fixed drug eruption must be considered. Acute hemorrhagic edema is a childhood form of leukocytoclastic vasculitis, which closely resembles EM minus. Erythema multiforme-like lesions in patients with systemic lupus erythematosus are known as Rowell's syndrome (▶ Chap. 54, "Lupus Erythematosus"). When mucosal involvement is present EM majus must be distinguished from Stevens–Johnson syndrome.

Histopathology

Initially, there is a lymphocytic perivascular infiltrate with edema and mild exocytosis of red cells. Eosinophils are not present. As the disease progresses, epidermal damage occurs. There is vacuolar degeneration of the basal layer and necrotic keratinocytes can be identified. In more severe cases, the entire epidermis may be necrotic. Subepidermal blisters result from the damage at the basement membrane zone. The presence of true vasculitis or neutrophils speaks against the diagnosis of erythema multiforme. Immunofluorescent studies have not identified any disease-defining antibodies.

Course

The outlook for EM minus is rapid resolution but with a high likelihood of recurrence. For EM majus, the opposite is true: a severe course but low likelihood (5%) of recurrence.

Therapy

The therapy of erythema multiforme depends on the suspected trigger and the disease severity. The uncomplicated course is self-limiting.

Topical Topical treatment is generally supportive and shows limited efficiency. Early, non-eroded target lesions can be treated with zinc lotion or topical corticosteroid creams or lotions, which may slow disease progression. In most cases erosions and crusts develop. They can be treated with wet soaks, ointments, and with antibacterial agents. Oral lesions may be treated with anesthetic gargles or rinses to make eating easier. Painful erosions of the lips are softened by oily or fat-moist measures.

Systemic If the erythema multiforme is severe, prednisolone 60–80 mg daily in tapered doses is effective in suppressing the cutaneous reaction, but is controversial because it may facilitate persistence of the triggering infection. Antiviral therapy is usually too late if started when lesions occur but very effective for prophylaxis. Long-term treatment with acyclovir or valacyclovir is recommended (▶ Chap. 9, "Human Herpes Viruses"). Patients with recurrent EM minus and no clinical signs of herpes simplex also usually benefit from prophylaxis.

In addition, various other immunosuppressive or immunomodulatory therapies have been successfully applied in severe long-term courses, including dapsone, hydroxychloroquine, azathioprine, methotrexate, intravenous

immunoglobulin (IVIG), and, increasingly, mycophenolate mofetil/mycophenolic acid.

2.4 Nodular and Plaque-Type Erythemas

The term nodular and plaque-type erythemas is not used much in English but is a mainstay of clinical diagnosis in Germany and a useful concept. The classic nodular erythema is erythema nodosum, the most common form of panniculitis (▶ Chap. 78, "Diseases of Fat"). There are a number of other disorders that present with acute, usually symmetrical red-brown nodules. Many are discussed below; others include leukemic infiltrates, lymphomas, Kaposi's sarcoma, and metastases. Multiple adnexal tumors, such as cylindromas, may look similar but are never as symptomatic.

These diseases are most likely reaction patterns, probably on the basis of a hypersensitivity reaction. The trigger factors of Sweet's syndrome are similar to those of erythema annulare centrifugum and erythema multiforme. Granuloma eosinophilicum faciei and erythema elevatum diutinum are less symptomatic and chronic variants of localized inflammation. All skin diseases listed here have in common that they show histological signs of leukocytoclastic vasculitis; however, they are not considered primary vasculitis.

2.4.1 Sweet's Syndrome

(Sweet 1964)

Synonym

Acute febrile neutrophilic dermatosis

Epidemiology

Sweet's syndrome is uncommon. Different types are distinguished according to the triggering factors. Classic Sweet's syndrome most often occurs in middle-aged women.

Etiopathogenesis

Initially, Sweet felt that acute febrile neutrophilic dermatosis was an allergic reaction to an infectious agent. The triggering by granulocyte colony-stimulating factor (G-CSF) and the overlap with other neutrophilic dermatoses (e.g., pyoderma gangrenosum) underlines the pathophysiological importance of neutrophils. In some cases, antibodies against antigens of neutrophils (ANCA) have been detected in the blood of patients.

Different forms are subdivided according to trigger factors:

- Idiopathic or classic form with no obvious cause; often begins in autumn; frequency 20–60% (later development of malignant disease has to be ruled out).
- Paraneoplastic Sweet's syndrome; Sweet's syndrome can reflect an underlying hematologic malignancy, usually myeloid or hairy cell leukemia, chronic lymphocytic leukemia, or solid tumors; frequency 10%. Atypical clinical forms are more commonly associated with hematological disorders. Elderly patients with anemia or thrombocytopenia without arthralgia are more likely to have malignancy-associated Sweet's syndrome.
- Infections (upper respiratory tract, gastrointestinal tract)/vaccinations: *Yersinia*, parvovirus B19, HIV).
- Drugs: G-CSF (most common), carbamazepine, chloroquine, azathioprine, clindamycin, hydralazine, oral contraceptives, minocycline, co-trimoxazole, imatinib, bortezomib.
- Pregnancy: self-limiting benign course in the first and second trimesters without increased fetal risk; recurrence is possible in subsequent pregnancies.
- Inflammatory systemic diseases and autoimmune diseases: chronic inflammatory bowel disease (most common), celiac disease, lupus erythematosus, Behçet's disease, Hashimoto thyroiditis, relapsing polychondritis.

A clinical and histological variant with high association to myelodysplastic syndrome is histiocytoid Sweet's syndrome. It is characterized by a dermal infiltrate mainly with lymphocytes and histiocytoid myeloperoxidase-positive cells.

Clinical Features

The clinical findings are impressive. The patient presents with fever, an elevated neutrophil count, and succulent, tender, dark-red plaques and nodules that can become confluent. The cutaneous findings are somewhat variable, ranging from the typical juicy plaques with pseudovesicles to erythematous patches to pustular lesions (Fig. 5). Rarely, bullous lesions are seen, more often associated with hematological malignancies. Another variant is acral and known as neutrophilic dermatosis of the dorsal hands. The skin lesions usually heal uneventfully, but may last for many weeks if not treated.

Extracutaneous involvement is known. Ocular involvement (conjunctivitis, episcleritis, iritis, ulcerative keratitis, choroiditis), rarely protracted lymphadenopathy, and splenic swelling (sterile splenic abscesses), very rarely serious lung involvement and vision loss due to occluding ocular venous vasculitis have been described.

Differential Diagnosis

The main criteria to be fulfilled are:

- A typical clinical picture of sudden succulent nodules or plaques
- Histology with papillary edema and massive neutrophilic infiltration

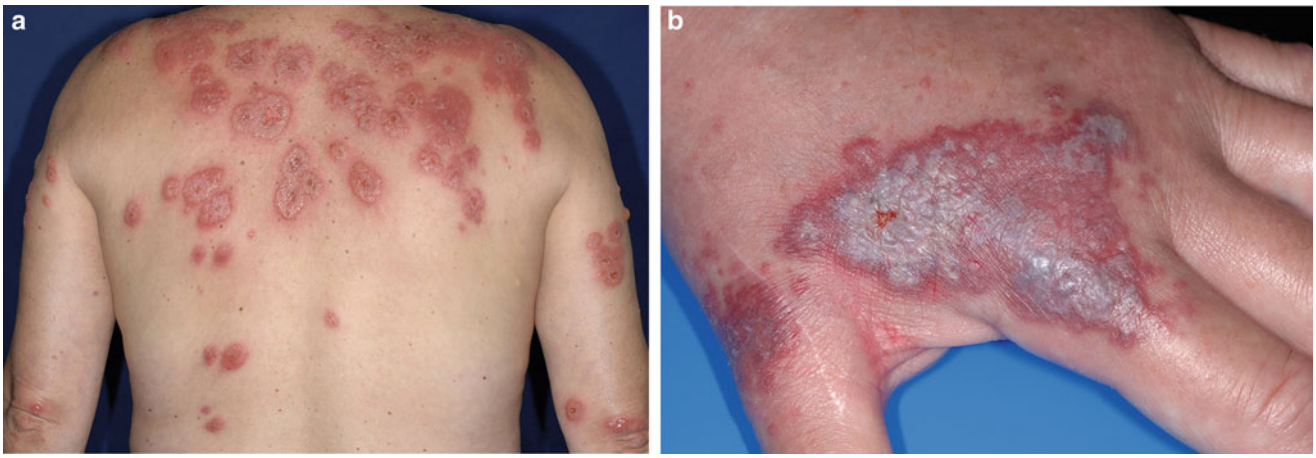


Fig. 5 (**a**) Sweet's syndrome. (**b**) Bullous Sweet's syndrome

Indicative criteria, of which at least two must be fulfilled:

- A previous infection or vaccination, lymphoproliferative or chronic inflammatory disease or pregnancy
- Fever >38 °C
- Erythrocyte sedimentation rate >20 mm/h, leukocytosis with a left shift
- A prompt response to systemic glucocorticoids

Erythema multiforme may be similar in appearance, but typically has iris lesions and more acral involvement. In addition, the neutrophil count is not elevated and the biopsy lacks neutrophils. Erythema elevatum diutinum is similar histologically but not clinically. Acute lupus erythematosus may present with infiltrated plaques; the serological evaluation and histology should help in the differentiation. An overlap between Sweet's syndrome and pyoderma gangrenosum is most often associated with myeloproliferative disorders, especially hairy cell leukemia. These patients have multiple, superficial, often bullous lesions; oral involvement and neutrophilic infiltrates in internal organs may be seen as well. Finally, an evolving leukemia or malignant lymphoma may be similar in appearance.

Histopathology
The main finding is an intense dermal neutrophilic infiltrate centered around vessels and often with nuclear dust. Whether or not the disease is primary vasculitis remains controversial. The upper dermis is edematous and the epidermis may show spongiosis and accumulations of neutrophils. Older lesions are dominated by lymphocytes. The vessel walls may be swollen, show fibrinoid degeneration, and reveal deposition of complement components and immunoglobulins with immunofluorescent staining. In histiocytoid Sweet's syndrome the infiltrate consists of histiocytoid immature myeloid cells, causing confusion with leukemic infiltrates.

Laboratory Findings
The erythrocyte sedimentation rate and the neutrophil count are elevated. The left shift may be significant, with 70–90% neutrophils and a total leukocyte count >20,000/mm^3.

Therapy
The underlying disease must be treated. The mainstay of symptomatic therapy is systemic corticosteroids. The initial dose used is 1 mg/kg bodyweight prednisolone daily, with tapering over 2–3 weeks. Antipyretics should also be prescribed. Antibiotics are of no benefit. In non-responsive or persistent cases, nonsteroidal anti-inflammatory drugs, colchicine (1–1.5 mg daily for 3 weeks), potassium iodide (600–1200 mg daily), dapsone, ciclosporin, or IVIG may be useful. Topical treatment is of little value, but corticosteroid creams or lotions or zinc lotion can be tried for early lesions.

2.4.2 Granuloma Faciale

(Wigley 1945; Pedace and Perry 1966)

Synonym
Granuloma eosinophilicum faciei

Epidemiology
Granuloma faciale is a disease of adults. Most patients are white men 40–60 years of age.

Etiopathogenesis
The cause is unknown. Early lesions sometimes show leukocytoclastic vasculitis, so that many consider granuloma faciale to be chronic or persistent vasculitis. Although an infectious etiology has often been suggested, no organism



Fig. 6 Granuloma faciale

has been reliably identified. Sun exposure has also been suggested as a trigger.

Clinical Features
Initially, one or more sharply delimited, minimally infiltrated nodules or plaques are seen. Sometimes the lesions are polycyclic. As they progress, they become more indurated, often with orange-peel surface markings. The typical color is red-brown (Fig. 6). Sometimes telangiectasias are observed. The sites of predilection are the cheeks, temples, nose, earlobes, and scalp, although any part of the face can be involved. Occasionally, extra-facial lesions are seen. Most lesions are asymptomatic, but occasionally burning or pruritus is described. Most patients are white adults. Spontaneous healing may occur, but more often the lesions are chronic. Rarely, there is extra-cutaneous involvement in the form of eosinophilic angiocentric fibrosis of the upper airways, a disease that can also appear without cutaneous findings.

Differential Diagnosis
Sarcoidosis, fixed drug eruption, lupus tumidus, lymphocytic infiltration, lymphoma or pseudolymphoma, and leukemia are the disorders to be excluded. Erythema elevatum diutinum is similar histologically but most often has smaller brown-red acral lesions.

Histopathology
Microscopic examination reveals a normal epidermis and a border zone of normal papillary dermis. At first there are accumulations of neutrophils and eosinophils around the vessels. Giant cells or foamy macrophages are uncommon. In very early lesions, leukocytoclastic vasculitis may be present. In more advanced lesions, fewer eosinophils are found and lymphocytes, plasma cells, and mast cells may be seen, as well as hemosiderin and fibrosis. Direct immunofluorescent examination may reveal nonspecific deposits of immunoglobulins and complement in the basement membrane zone.

Therapy
Topical

Intralesional corticosteroids are probably the best first approach. Corticosteroid creams and ointments rarely produce improvement. Tacrolimus or pimecrolimus can be an alternative to intermittent glucocorticoid therapy. Intradermal injection of glucocorticoid crystal suspensions may also be effective, but, because of the facial location of the target lesions, it should be used with care. A wide variety of destructive methods, including cryotherapy and pulsed dye (585 nm) or CO_2 laser therapy, produce good but often not lasting results.

Systemic Due to the benign nature of the disease, the therapeutic risk should be considered. Dapsone, chloroquine and hydroxychloroquine are particularly recommended.

2.4.3 Erythema Elevatum Diutinum

(Hutchinson 1878; Bury 1889; Crocker and Williams 1894)

Synonym
Extracellular cholesterolosis (Urbach et al. 1932)

Epidemiology
This rare disorder shows no sex preference. Adults between the third and sixth decades of life are mostly affected, but individual cases in children have also been described.

Etiopathogenesis
The cause is unknown. The basic process may be chronic vasculitis, similar to granuloma faciale; some consider the two disease variations to have a common background. Immunological markers for streptococcal or other bacterial antigens are often present. Erythema elevatum diutinum has been described in association with Crohn's disease, collagen-vascular disorders, gout, paraproteinemia, antithrombin III deficiency, and even HIV/AIDS or a suspected drug reaction. In patients with erythema elevatum diutinum isolated ANCAs were detected.

Clinical Features

The most common sites are the extremities, especially the knees, elbows, and backs of the hands and feet (Figs. 7, 8, and 9). Multiple papules and nodules are present in a symmetric fashion. Occasionally, larger, sometimes annular plaques develop with a central depression. Although early lesions tend to be red, later on they acquire red-brown tones. Hemorrhage and ulceration are extremely uncommon. The lesions are often painful and heal with hyperpigmentation. Some patients have arthralgias; peripheral ulcerative keratitis has been reported, but other internal problems are rare. Vasculitis does not seem to occur in other organs. The disease course is chronic, over years; individual lesions may disappear as new ones develop.



Fig. 7 Red purple nodes in the first stage of erythema elevatum diutinum



Fig. 8 Nodular erythema elevatum diutinum on the dorsal feet and toes

Differential Diagnosis

Lesions are often misdiagnosed clinically as granuloma annulare (▶ Chap. 42, "Granulomatous Diseases"), but the individual lesions are much more succulent and red-brown; sometimes, lesions on the legs are confused with dermatofibroma.

Histopathology

Microscopic examination reveals a diffuse dermal infiltration of neutrophils and eosinophils. The dermal blood vessels are damaged with fibrin deposits, leukocytoclasia, and nuclear dust. Older lesions may be more granulomatous or even fibrotic. Necrobiosis is not seen. A histological variant is extracellular cholesterolosis (Urbach et al. 1932), in which there are cholesterol clefts and a granulomatous response. Despite these findings, abnormal lipid metabolism has not been identified.

Therapy

Dapsone is the single most effective agent in doses of 50–150 mg daily. Penicillin can be helpful if a streptococcal



Fig. 9 Patchy lesions of erythema elevatum diutinum on the ear helix

infection is identified. Systemic corticosteroids may also be tried. Intralesional corticosteroids can be used to treat individual painful lesions.

2.5 2.4. Necrolytic Acral Erythema

(El Darouti and Abu el Ela 1996)

This peculiar localized eruption is almost a specific marker for hepatitis C infection. About 3% of the world's population is infected with this virus, and are thus at an increased risk of chronic hepatitis and hepatocellular carcinoma. The skin eruption was first described in Egypt and has been reported primarily in the Middle East but also in the USA and elsewhere. The patients have intensely pruritic, acral, erythematous dusky plaques with adherent scale. The lesions slowly extend but do not involve the palms and soles. On histology, early lesions resemble eczema with moderate acanthosis and variable spongiosis and inflammation. Fully developed lesions show a psoriasiform epidermal hyperplasia with marked papillomatosis. Associated findings include parakeratosis, focal hypergranulosis, subcorneal pustule formation, and epidermal pallor. A typical sign is keratinocyte necrosis, with involved areas sometimes becoming confluent in the upper epidermis and/or tracking along the acrosyringia, dilated vessels, and the dermal inflammatory infiltrate. Pigment incontinence is seen at all stages.

Therapy

The lesions are refractory to topical therapy, but sometimes improve when the underlying hepatitis C infection is treated with interferon and antiviral agents. Success with oral zinc has also been described.

3 Papulosquamous/Erythematosquamous Diseases

The German term for this group of disorders is erythematosquamous (instead of papulosquamous) diseases, which is more accurate since the unifying clinical features are erythema and scaling, rather than papules. In 1902, Brocq grouped some of these skin diseases together under the term **parapsoriasis** owing to their clinical similarity to psoriasis, including **pityriasis lichenoides** (parapsoriasis guttata), **parapsoriasis en plaques**, and **parakeratosis variegata**. This classification is outdated today. The forms en petites plaques and en grandes plaques, which are still called parapsoriasis today, and their variant, parakeratosis variegata, are now better considered to be cutaneous T cell lymphomas (▶ Chap. 110, "Cutaneous Lymphomas"). Pityriasis lichenoides, pityriasis rubra pilaris, and other diseases listed here have in common that they have a typical bran-shaped scale. However, they can be distinguished from each other as independent entities based on the clinical picture and course. An overview is given in Table 3.

Table 3 Age at first manifestation and natural course of selected erythematosquamous diseases

Disease	Onset of disease (years)	Course
Pityriasis rosea	10–40	Weeks
Pityriasis lichenoides acuta	8–16	Months
Pityriasis lichenoides chronica	20–40	Years
Pityriasis rotunda	20–50	Months to decades
Pityriasis rubra pilaris	Newborns up to old age	Months to decades
Psoriasis	Children up to old age	Lifelong
Reiter's disease	20–40	Months
Lichen planus	30–60, also in children	Years
Parapsoriasis en plaques	Very rare in childhood	Decades

3.1 Pityriasis Rosea

(Gibert 1860)

Epidemiology

Pityriasis rosea is a common disease of adolescents and young adults, accounting for 1–2% of dermatology outpatient visits. It tends to appear in cycles and is more common in the fall and winter months. Mini-epidemics can be seen among those with close contact, such as family members, classmates, or fellow workers. Most cases occur between the ages of 10 and 35.

Etiopathogenesis

Pityriasis rosea has long been assumed to represent a delayed reaction to a viral infection. The seasonal clustering and occurrence among family members or classmates point in this direction**.** Serological investigations and the detection of specific DNA indicate an infection with human herpes viruses types 6 and 7 and possibly also type 8 is a trigger in some patients. An association with the H1N1 virus has also been reported. Connections to other viruses such as Epstein–Barr virus or cytomegalovirus, or a streptococcal infection could not be proven. Immunological studies show an infiltration with $CD4^+$ T cells and a proliferation of epidermal Langerhans cells associated with elevated peripheral levels of interleukin-22, a product of Th17 cells that plays a role in

cutaneous antimicrobial immune responses and also in psoriasis.

Clinical Features

The clinical picture is quite distinct. Most patients can be diagnosed at a glance. The initial lesion, also known as the herald patch, is usually on the trunk. It is larger than the subsequent lesions, usually several centimeters at its widest diameter. The lesion is initially red, but later pale pink with a collarette of scale that typically points centrally. The central part of the lesion is covered with fine scale and may appear atrophic or depressed. This initial lesion may be overlooked by the patient or diagnosed and treated as a dermatophyte infection.

Days to weeks later, the disease starts to spread, alarming the patient and usually prompting the first office visit. Many smaller oval scaly lesions are seen often following the skin lines on the trunk, producing a Christmas tree pattern. The individual lesions are very similar to the herald patch but always smaller. The trunk, abdomen, and neck are the most frequent sites; the extremities and face are rarely involved. The lesions typically do not itch or are only minimally symptomatic. No mucosal lesions develop. Only occasionally do patients report fever, chills, or malaise during the early stages.

The disease may take a slightly different course in Black and Hispanic patients. Here, the neck, face, and distal aspects of the extremities are prominently involved; this pattern has been called inverse pityriasis rosea. The lesions tend to be much smaller, are often follicular, typically very pruritic, and heal with significant postinflammatory hypopigmentation. In smaller children, the lesions may have the same pattern but are usually more urticarial in nature. Other, even less common variants include hemorrhagic and vesicular forms (Fig. 10).



Fig. 10 Pityriasis rosea

Differential Diagnosis

Secondary syphilis can appear very similar; a serological test for syphilis should be recommended to every pityriasis rosea patient. Tinea corporis can usually be excluded by the pattern, whereas the negative KOH examination is confirmatory. Early cases of psoriasis may be similar in appearance, but should evolve differently. Petaloid seborrheic dermatitis has similar scaly lesions, involving the neck, axillae, and sternal region. Collarette scale is not seen. The disease is more chronic and usually there are other signs of seborrheic dermatitis present on the scalp or face. Drug eruptions may also mimic pityriasis rosea. The most classic agents are the gold salts, but captopril, isotretinoin, ketotifen, and metronidazole have also been incriminated.

Histopathology

The epidermis shows minimal, often focal, spongiosis and tiny parakeratotic accumulations or caps. In the dermis, there is a sparse perivascular infiltrate. Occasionally, there is exocytosis of red cells.

Course

The disease usually persists for 3–6 weeks but may last for 3–4 months, generally in patients who have been promised that the rash will disappear promptly. Healing may occur with both postinflammatory hyperpigmentation and hypopigmentation. Recurrences are uncommon ($\leq$3%): **pityriasis rosea recidivans**.

Therapy

It should be explained to the patient that treatment will play only a small role in how rapidly the lesions disappear. Most patients are satisfied once they learn that the disease is harmless and self-limited, even though it looks so worrying. Patients should be encouraged not to wash frequently, as this leads to dryness and secondary irritation. They should be encouraged to lubricate their skin well. Topical corticosteroids or UVB/UVA1 radiation can be used to help relieve pruritus.

Systemic antihistamines can be used for severe pruritus. Systemic corticosteroids are sometimes employed, especially in Black patients with severe disease and marked hypopigmentation, but their benefit is unproven. Aciclovir 3× 400 mg/day over 1 week seems to improve disease symptoms.

3.2 Pityriasis Rubra Pilaris

(Devergie 1856)

Epidemiology

Pityriasis rubra pilaris (PRP) can appear in any age group, but follows a bimodal distribution, involving children and adults. It has a prevalence of 2/100,000 inhabitants and is about 1000 times less common than psoriasis.

Etiopathogenesis

The cause of pityriasis rubra pilaris is unknown. Occasional familial cases have been described, especially when the onset is in childhood, but most are sporadic. Autosomal-dominant inheritance, autosomal-recessive, and X-chromosomal cases have been reported. In some patients a *CARD14* mutation could be detected. These individuals develop erythematosquamous lesions (also termed CAPE, for CARD14-associated papulosquamous eruption) with features of psoriasis and/or PRP.

Several cases have been reported among HIV/AIDS patients, but PRP is not usually associated with immunosuppression. In other instances, it may follow a severe infection (EBV, CMV, VZV, hepatitis, and HIV infection). Although PRP shares some features with psoriasis, it is a distinct disorder, probably more closely related to disorders of keratinization than to inflammatory processes. Cases of PRP associated with malignancies and autoimmune diseases have been described.

Clinical Features

Pityriasis rubra pilaris is characterized by follicular erythematous papules and more diffuse erythematous patches with pityriasiform scaling. It has a variety of clinical forms (Table 4). The disease often smolders along in a relatively uncharacteristic pattern, causing the patient distress and diagnostic confusion until the classic findings develop, which makes the correct diagnosis easier. Often, the disease process begins on the scalp with erythema and fine white scaling. Facial patches may be deep-red with thick crusts, especially in the nasolabial folds and eyebrows. On the body PRP starts as erythematous to salmon-colored patches that become papular and lichenified. Typically, areas of normal skin are spared (nappes claires; Fig. 11). Predilection sites are follicular orifices on the back of the fingers and hands, extensor

Table 4 Classification of pityriasis rubra pilaris; after Griffiths (1976) and (1992)

Type		Percentage of all cases	Clinical features and course
Adult			
I	Classic	55	Abrupt onset, classic clinical picture, lasts 2–5 years
II	Atypical	5	Slow onset, localized lesions, alopecia, chronic course
Childhood			
III	Classic	10	Same as adult classic version but resolves in 1–2 years
IV	Circumscribed	25	Association with infections possible; psoriasis-like lesions over joints, first onset before puberty, course over months to years
V	Atypical	5	Limited to palms and soles, onset in the first year of life, association with *CARD14* mutation suspected
Other			
VI	HIV-associated		Similar to type II



Fig. 11 White-spread erythematous pityriasis rubra pilaris lesions around islands of normally appearing skin (so called *nappes claires*)



Fig. 12 Follicular and confluent lichenificated lesions of pityriasis rubra pilaris with orange-brownish color aspect



Fig. 13 Typical palmar involvement in pityriasis rubra pilaris with orange-brownish infiltrate involving the whole palm and hyperkeratosis

surface of the limbs, areas above the large joints, the chest, abdomen, and buttocks (Fig. 12). Other common sites are the palms and soles, which are initially red and then become hyperkeratotic and fissured, often with an orange hue (Fig. 13). The disorder is frequently pruritic.

When one looks closely, many of the individual lesions are small papules, usually centered around follicles. They are especially prominent on the fingers, where they have been described as nutmeg-grater lesions. The papules are covered by a keratotic cap or dell and may contain a broken-off hair. Sometimes the papules coalesce to produce a plaque; conversely, sometimes they develop in erythematous areas. Nail changes may also be seen, including subungual hyperkeratoses, nail plate thickening, and discoloration (Fig. 14); pits, oil drop discoloration, and other classic changes of nail psoriasis are not found. Oral lesions do not occur. PRP may rapidly evolve into an erythroderma. Usually, at least a few areas are spared; follicular papules and palmoplantar involvement indicate the correct diagnosis.

In children, the disease usually starts slowly. About half of childhood patients have circumscribed PRP with persistent papules and erythematous patches over the knees and elbows. These children are often misdiagnosed as having psoriasis. In the other variants, the classic features are present in limited or modified form. For example, in the childhood atypical or type V variant, the lesions on the palms and soles resemble those of classic PRP. Whether there are any other unifying features awaits molecular biological investigation of these rare problems.

Differential Diagnosis

The differential diagnosis is discussed under the clinical features. In our experience, PRP is frequently misdiagnosed. Early cases often resemble seborrheic dermatitis or psoriasis; the uncommon follicular variant of psoriasis is very similar. In addition, patients are seen who have both PRP and psoriasis. In children, the usual working diagnosis is atopic eczema. When the palm and sole involvement is prominent, palmoplantar keratoderma or another disorder of keratinization may be considered. The spiny keratotic papules may look lichenoid, but lack a violaceous color and are not seen in the typical sites for lichen planus. If one keeps looking for areas that have been spared, salmon-colored



Fig. 14 Extensive subungual hyperkeratosis in pityriasis rubra pilaris with nail involvement

lesions, and palmoplantar disease, some initially puzzling papulosquamous disorders will eventually be diagnosed as PRP.

Histopathology
Microscopic examination reveals acanthosis, a thickened granular layer, and orthokeratosis alternating with parakeratosis, both vertically and horizontally. Shoulder parakeratosis is often found at the openings of follicles or actually plugging them. There may be spongiosis, especially over the tips of the dermal papillae. In the dermis, there is a lymphocytic perivascular infiltrate with few diagnostic features. The keratotic papules typically have lamellar follicular keratinization at the level of the sebaceous duct. A biopsy usually allows psoriasis to be excluded.

Course
In children with classic disease, the lesions tend to clear rapidly with treatment, often within 1 year, whereas adults usually respond over a period of several years.

Therapy
Topical

Exogenous irritating factors should be avoided, as the skin is easily irritated in PRP, leading to iatrogenic flares. For this reason, both phototherapy and photochemotherapy are of limited value. Finding the best emollient for a given patient is very important in caring for this most recalcitrant disease. Topical corticosteroids, either high-potency or under occlusion, may produce some improvement and reduce the amount of systemic therapy needed. Additionally, positive effects of topical calcineurin inhibitors have been reported. Exacerbation has been observed under imiquimod.

Phototherapy/Systemic In contrast to psoriasis, PRP usually does not respond to phototherapy. Acitretin in a dosage of about 0.5 mg/kg bodyweight can be given. The anti-inflammatory effect is often not sufficient. Years of therapy are often required; thus, skeletal and triglyceride monitoring is essential, in addition to the usual pregnancy precautions. Alternatives are methotrexate and ciclosporin, both used in doses similar to those for the treatment of psoriasis. PRP does not respond to either agent as quickly as psoriasis does. Additionally, alitretinoin can be used. A large number of case reports prove the good effects of therapy with TNF inhibitors (infliximab, adalimumab, etanercept), partly in combination with low-dose methotrexate or retinoids. Adalimumab and etanercept can also be used analogous to the treatment of psoriasis in the severe refractory forms of childhood. The IL-12/-23p40 antagonist ustekinumab showed good efficacy in individual cases. IL-17A antagonists are also effective in PRP. Patients with HIV infection often show improvement after successful antiviral therapy, possibly in combination with retinoids.

3.3 Pityriasis Lichenoides

Pityriasis lichenoides refers to two diseases that were for years felt to be separate entities but today are considered part of the same spectrum. They are pityriasis lichenoides et varioliformis acuta (PLEVA) (Mucha 1916; Habermann 1925) and pityriasis lichenoides chronica (PLC) (Jadassohn 1894; Juliusberg 1899). Although each will be considered separately we feel that they belong together because of clinical and histological similarities and because of frequently overlapping cases, which begin as pityriasis lichenoides et varioliformis acuta and end up as pityriasis lichenoides chronica.

Epidemiology
Both diseases occur in childhood with peaks in preschool (2–3 years) and primary school (5–7 years) as well as in adults (mean age about 40 years). The prevalence is about 20/100,000 inhabitants. Infantile pityriasis lichenoides shows a more extensive involvement, with more frequent

recurrences, a stronger tendency toward postinflammatory pigment shift and a lower response to usual therapy concepts. Men are more commonly affected than women.

Etiopathogenesis
Three theories are discussed:

- **Infection-triggered immunological reaction:** about 20% of cases with infantile pityriasis lichenoides are preceded by an infection, usually of the upper respiratory tract (pharyngitis), chickenpox, or gastroenteritis. The following pathogens are discussed as trigger factors: *Toxoplasma gondii,* Epstein–Barr virus, HIV, cytomegalovirus, parvovirus B19, *Staphylococcus aureus*, and β-hemolytic streptococci.
- **Inflammatory lymphoproliferative disease:** the target site is the superficial dermal vessels, which are damaged by a true lymphocytic vasculitis. In some cases, a clonal expansion of T cells can be identified (CD5, CD7, $CD62L^-$; $CD30^+$). PLEVA and **f**ebrile **u**lceronecrotic **M**ucha–**H**abermann **d**isease (FUMHD) can be associated with $CD8^+$ cytotoxic cells, whereas in PLC rather $CD4^+$ cells dominate. $CD8^+$ cells play a role in controlling clonally expanding $CD4^+$ cells and in damaging epidermal cells. A transition of pityriasis lichenoides into a cutaneous lymphoma has been rarely described.
- **Immunocomplex hypersensitivity vasculitis:** this concept is based on the detection of immunoglobulins and complement in some immunofluorescence studies, not only in the junction zone but also in the wall of dermal vessels. It partly overlaps with the first theory. The detection of chimeric maternal keratinocytes in some patients with pityriasis lichenoides is conspicuous; thus, a GVHD-like reaction directed against epidermal cells could also be involved.

Clinical Features
Pityriasis lichenoides et varioliformis acuta most often starts on the trunk with erythematous papules that continuously erupt over a period of weeks, developing scaling and crusts, and often ulcerating. The lesions may be hemorrhagic and vesicular (Fig. 15). They typically resolve with subtle scars. For all these reasons the tongue twister of a name was coined: the lesions have scales (pityriasis), may be papular (lichenoides) and erythematous, heal with scars (varioliformis), and appear suddenly (acuta). As a memory hook, pityriasis lichenoides et varioliformis acuta has been described as looking like chronic chickenpox. Most patients are asymptomatic, often surprisingly as they look as though they should be sick. There is no mucosal involvement. The average duration of infantile pityriasis lichenoides is about 2 years. However, disease activity can last up to 10 years. A relapse rate of 80% is described.



Fig. 15 Pityriasis lichenoides et varioliformis acuta

Several dozen cases of severe PLEVA have been described in the literature. Particularly children, adolescents, and young adults were affected; the mean age was 27 years, and twice as many men as women were affected. Sometimes, ulcerations dominate the picture. They may develop spontaneously or in an already developed case of pityriasis lichenoides et varioliformis acuta. The ulcerations may be hemorrhagic, painful, and associated with fever. This constellation is known as **FUMHD** (Degos et al. 1966). In many cases, bacterial superinfections occur with hypertrophic scars after healing. Oral, conjunctival, or genital mucosa is often involved. In addition, systemic manifestations are frequently observed, including gastrointestinal, neurological or pulmonary involvement, cardiomyopathy, rheumatological complaints, or coagulation disorders. Some patients show megaloblastic anemia or pancytopenia. Cardiovascular mortality is very high at 20%; almost all were older adults.

Pityriasis lichenoides et varioliformis acuta can develop into **PLC**, which explains the coexistence of typical clinical signs of PLEVA and PLC in about 10% of patients with

infantile pityriasis lichenoides. PLC can also develop primarily. As the disease is almost always asymptomatic, the patient usually presents with a rash in full bloom. The clinical picture is polymorphic, as new lesions appear while old ones begin to fade. The trunk and proximal parts of the extremities are the main sites of involvement. The initial lesions are 2–10 mm dome-shaped red or red–brown papules whose surface is either dull or has a slight shimmer. Although there is a small amount of fine scaling, it is only noticed when the lesion is scratched. Hemorrhagic lesions occur but are uncommon. The papules tend to expand as they become flatter and often either lighter or browner. They also develop a more prominent compact scale that covers them. After several more weeks, the lesion is entirely flat and has left behind the classic clinical sign of pityriasis lichenoides chronica – a thick, sharply defined, single large scale. When this scale is peeled away, normal skin is revealed. The scale is described as *Oblaten*-like in German. *Oblaten* refers to the paper-thin wafers used in religious ceremonies. In darker-skinned individuals, pityriasis lichenoides chronica often heals with hypo- and hyperpigmentation. The disease is chronic for a few weeks, several months, or even years (Figs. 16 and 17).

Differential Diagnosis

The differential diagnosis of PLEVA is limited. The lack of fever and oral involvement contraindicates varicella. In the early stages, many other papulosquamous disorders can be considered, such as guttate psoriasis and secondary syphilis, but the hemorrhage and crusts soon make the diagnosis clearer and the histology confirms the clinical impression.

Although the initial lesions of PLC may resemble guttate psoriasis, the monomorphous nature of psoriasis usually allows a rapid distinction. Lichen planus is also a more uniform eruption and has far less scaling, as well as frequent oral involvement. Secondary syphilis can also be similar in appearance, but is more acute and monomorphous.

Histopathology

Pityriasis lichenoides et varioliformis acuta is the prototypic disease featuring lymphocytic vasculitis. There is endothelial cell swelling and exocytosis of erythrocytes and lymphocytes, but few neutrophils and no nuclear dust. No fibrin deposition is seen. There are large accumulations of lymphocytes, primarily suppressor T cells, around the vessels. Some of the erythrocytes may reach the epidermis, where reticular degeneration and necrosis may occur. Occasionally, there may be clinically typical cases of pityriasis lichenoides et varioliformis acuta that, on biopsy, show large $CD30^+$ lymphocytes. Such cases overlap with lymphomatoid papulosis. In FUMHD the histological changes are very pronounced, especially the signs of cytotoxic epidermal cell damage. Some authors also reported moderately prominent signs of leukocytoclastic vasculitis. In PLC, lymphocytic vasculitis may be present, but it is much less prominent. Instead, the lymphocytes tend to be perivascular or form a lichenoid pattern along the dermoepidermal junction. There may be spongiosis, minimal acanthosis, and parakeratotic scale. Rarely, accumulations of lymphocytes can be found in a parakeratotic horny layer. Exocytosis of erythrocytes also occurs. In later lesions, the infiltrate is minimal and only the thick, compact scale is seen.

Therapy

The treatment depends on clinical manifestation, age, concomitant diseases (e.g., infection), and the course. In the case of mild illness, especially in childhood, it is often possible to wait owing to a good prognosis. The febrile ulceronecrotic variant should be regarded as a dermatological emergency that has to be treated rapidly and aggressively under inpatient conditions.



Fig. 16 Phenotypic transition from PLEVA to PLC with clinical signs of both conditions



Fig. 17 PLC lesions reminiscent of Pityriasis rosea with collarette scaling

Topical External therapy is strictly symptomatic and rarely useful. Topical corticosteroids are frequently recommended, but they do little to influence the disease course.

Phototherapy/Systemic Phototherapy is often the treatment of choice for PLC. UVB 311 nm, oral PUVA, or bath PUVA therapy can all be tried. One must always discuss with the patient the possibility of triggering a disease flare with phototherapy. Climate therapy, as discussed under psoriasis, with sun exposure by the sea or at higher elevations is also a possibility. In both diseases systemic antibiotic therapy is frequently recommended and reasonable to try. Many patients improve with no therapy and others continue to worsen under antibiotic treatment. Usually, tetracycline (1.0–2.0 g daily), doxycycline (100–200 mg daily), erythromycin (1.0 g daily), or azithromycin (250–500 mg daily) is recommended. Treatment should be carried out for 3–4 weeks. It can be combined with a short course of systemic corticosteroids. Dapsone also has its proponents. Very low-dose methotrexate therapy can be amazingly effective, but it is rarely used because of the self-limited nature of the disease and the age of the patients. Alternatives include ciclosporin and retinoids. None is very effective and we tend to avoid them.

In FUMHD, immunosuppressive therapy with methotrexate and high-dose systemic glucocorticoids is indicated. Patients suffering from FUMHD showed elevated TNF-α levels. Consequently, a high response to TNF inhibitors was detected. However, there are also case reports of pityriasis lichenoides manifesting during ongoing therapy with TNF inhibitors. In some cases, IVIG and extracorporeal photopheresis were successfully applied.

3.4 Pityriasis Rotunda

(Matsuura 1906; Toyama 1906)

Epidemiology
Pityriasis rotunda is an uncommon disorder in Western countries. Patients typically have multiple, large, scaly round or oval patches on their trunk or buttocks that become hyper- or hypopigmented. Most patients are Asian or Black. In these groups, the lesions are usually few in number and generally hyperpigmented. There is usually no familial tendency; instead, pityriasis rotunda may be a marker for an underlying malignancy or chronic illness. In whites, it is often familial, usually hypopigmented, and not associated with underlying diseases. Familial cases have also been described in Asians. Patients are between 20 and 45 years old. Men are affected as often as women.

Etiopathogenesis
The cause is not exactly known. The microscopic picture is not helpful, usually showing modest hyperkeratosis and varying pigmentary changes. It seems that the familial cases may represent an uncommon form of congenital ichthyosis. One could argue that the sporadic form represents a variant of acquired ichthyosis, which can also be a marker for systemic diseases. Early acquired cases must be separated from tinea corporis and large-patch parapsoriasis. About 10% of patients have carcinoma of the lung, liver, and stomach. About a third have other diseases such as tuberculosis, malnutrition, cirrhosis of the liver, and leukemia. Based on the cases described, the following classification has been proposed:

- Type I: Black and Asian patients with <30 hyperpigmented lesions without familial accumulation; in 30% an association with a malignancy or systemic disease is found.
- Type II - white patients with >30 mostly hypopigmented lesions, occasionally with familial accumulation, without associated disease.

However, since the introduction of the classification in 1994, cases with characteristics of both types have been described.

Clinical Features
The typical lesions are very variable in their expression; in some patients there are only a few lesions, in others more than 100. They measure a few centimeters to over 20 cm. The disease can become self-limiting after months. In other cases, the course of disease can be chronic for years or decades. Episodic courses with exacerbation in winter have been described.

Differential Diagnosis
Pityriasis versicolor, pityriasis rosea, pityriasis alba, nummular eczema, parapsoriasis, erythrasma, leprosy, fixed drug eruption have to be considered as clinical differential diagnoses.

Histopathology
The histological picture is not specific. Hyperparakeratosis with a reduced or missing granular layer, mild spongiosis and a superficial, perivascular accentuated lymphohistiocytic infiltrate are often observed. Changes in pigmentation (increased basal pigmentation, pigment incontinence) can be seen.

Therapy
Treatment is symptomatic, with emollients. In some cases, 10% urea creams used along with topical tretinoin have cleared the condition. Successful treatment of the underlying disease can also lead to an improvement in pityriasis rotunda. In familial cases, an improvement was described with entry into puberty.

4 Erythroderma

(Hebra and Kaposi 1868)

Synonym
Exfoliative dermatitis

Definition
Erythroderma refers to skin that is diffusely red and inflamed with varying degrees and types of infiltration and scaling. More than 90% of the skin is involved, from head to toe.

Epidemiology
Few data are available on erythroderma. There is a pronounced male predominance (2–4:1) which is even higher among adults. When children are excluded from the statistics, the average age is over 60 years.

Etiopathogenesis
The major causes, each accounting for about 25% of cases, are drug reactions, psoriasis, and atopic eczema, with mycosis fungoides present in 10%. Table 5 lists the likely causes of erythroderma in the different age groups. The most useful classification is:

- Primary erythroderma: here, the changes develop on normal skin. Typical causes are drug eruption, mycosis fungoides, Sézary's syndrome, and, less often, as a paraneoplastic marker for other malignancies.
- Secondary erythroderma: in this case, the erythroderma represents the worsening and spreading of an existing skin disease. Typical causes are atopic eczema in children, psoriasis, and PRP in adults, and seborrheic dermatitis in the elderly. Often, the erythroderma is so extensive that the underlying disease is not readily apparent.

Table 5 Likely causes of erythroderma in different age groups

Age group	Diseases
Newborns	Ichthyosis (usually lamellar or bullous ichthyosiform erythroderma) Comèl–Netherton syndrome Staphylococcal scalded skin syndrome Immunodeficiency disorders
Children	Atopic eczema Pityriasis rubra pilaris Psoriasis Staphylococcal scalded skin syndrome
Adults	Drug reactions Psoriasis Atopic eczema Hematological malignancies (mycosis fungoides, Sézary's syndrome) Pityriasis rubra pilaris Lichen planus Pemphigus foliaceus Crusted scabies
Elderly	Idiopathic erythroderma (red man syndrome) Seborrheic dermatitis

Erythroderma has profound effects on the entire body. The widespread inflammatory response, increased blood flow, and marked desquamation all take their toll. The skin is usually infiltrated by a wide range of inflammatory cells that release a barrage of cytokines, affecting virtually every organ. The vasodilatation and increased blood flow lead to chills and impaired temperature control. Hypothermia is a possible complication, especially when the patient is placed in an air-conditioned hospital room. Increased evaporation made possible by increased blood flow and damaged epidermal barriers leads to dehydration and fluid problems. The basal metabolic rate is increased, often then calling into play secondary endocrine control mechanisms. Protein loss in the form of desquamation and exudation is significant, resulting in hypoproteinemia. Scales alone may account for up to 10 g/m^2 of protein loss daily. Typically, there is hypoalbuminemia with a relative increase in immunoglobulin, especially γ-globulins. Protein-losing enteropathy may also lead to reduced levels of iron, folic acid, and other vitamins, often producing a profound anemia. Cardiovascular, renal, and hepatic complications are common. The hair and nails are influenced by the increased demands; thus, telogen effluvium and transversal nail bands are expected. In more severe cases, anagen effluvium and even acute nail loss may occur.

If the erythroderma persists for any period of time, the patient's general condition will decline severely. In the past, patients often died from erythroderma. Today, far more advanced supportive care and, to a lesser extent, better diagnosis and care for the underlying skin condition, have made this a rarity. Drug-induced erythroderma usually improves dramatically when the causative agent is identified and stopped. Occasionally, it may evolve into toxic epidermal necrolysis, but even such patients may do well with supportive care. The prognosis in cases associated with malignancies tends to follow the underlying disease. In patients with erythroderma associated with a chronic skin disease, such as psoriasis, the course tends to be chronic. These patients improve but often suffer relapses and often go for months or years on the verge of recurrent erythroderma. Clearly, the prognosis also depends on the cause of the erythroderma.

Clinical Features
The clinical diagnosis of erythroderma is usually straightforward. Erythroderma refers to skin that is diffusely red and inflamed, with varying degrees and types of scaling. The nose and paranasal areas may be spared (**nose sign**). Acute erythroderma may feature lamellar scaling whereas chronic erythroderma is more likely to be psoriasiform or pityriasiform. The skin is often infiltrated as well.

Erythroderma should not be diagnosed unless more than 90% of the skin is involved from head to toe. Pruritus is common and often unbearable. Small areas of sparing can be a valuable clue to diseases such as mycosis fungoides or PRP (→ nappes claires).

Diagnostic Approach
The trigger of the disease is the decisive issue. One must try to distinguish between primary and secondary erythroderma. Finally, there are cases in which neither an underlying skin disease nor an exogenous trigger is identified. The points described below should be considered when trying to narrow down the causes.

History The existence of a previous skin disease and how it was treated are crucial factors. In addition, a careful drug and general history is needed.

Clinical Features One must search carefully for subtle signs of an underlying disease, such as gluteal cleft, nail, or scalp involvement in psoriasis, various signs in atopic eczema, oral involvement in lichen planus, or areas that have been spared in PRP.

Histopathology If possible, histopathology should be obtained before starting treatment, especially with glucocorticoids. One or even several skin biopsies should be taken. Although they sometimes solve the puzzle, they may just as well be non-diagnostic. Usually, histology will allow mycosis fungoides to be ruled out. Epidermal necrosis should suggest early, perhaps unexpected toxic epidermal necrolysis. Specific clues to psoriasis, atopic eczema, or other dermatoses are usually blurred by the inflammation. One finds spongiosis, lymphocytic perivascular inflammation, and edema. Later on, parakeratosis or scaling may be seen.

Lymph Node Evaluation Many patients have lymphadenopathy, which, although usually reactive or dermopathic, may reflect an underlying lymphoma. Most commonly, the axillary and inguinal nodes are involved. If sonography does not allow a clear diagnosis of inflammatory lymphadenopathy, a lymph node biopsy should be performed to exclude a lymphoma.

General Evaluation The patient should be carefully evaluated for underlying disease. Useful approaches include chest X-ray, computed tomography of the abdomen, pelvis, and thorax, routine blood evaluation, iron and folic acid levels, immunoelectrophoresis, and perhaps bone marrow examination. The presence of a pre-existing dermatological condition obviously narrows the search and allows many steps in the diagnostic workup to be avoided.

Course
The only truly acute erythroderma is that associated with drug eruptions. It overlaps with severe skin reactions and is potentially fatal, but once the offending agent is identified and avoided, clearing is rapid and recovery complete. In all other forms of erythroderma, the tendency is toward chronicity, and the course of the underlying disease determines that of the erythroderma.

The most common causes of death in erythrodermic patients are pneumonia, sepsis, and cardiovascular diseases. The elderly often show infections, electrolyte disorders, and cardiac failure. Global mortality rates range from 4.6% in industrialized countries to 64% in developing countries. The prognosis is improved by rapid diagnosis and adequate therapy.

Therapy
The treatment of erythroderma can be complex. Patients may require fluid restoration, parenteral nutrition, vitamin supplements, and should be carefully monitored for renal or cardiac failure. If the skin is generally and severely affected, patients should be treated in a similar manner to burn victims and transfer to a burns unit may be considered. In the case of an underlying malignancy, this must be treated. The key treatment approach will be derived from the identification of the underlying skin disease, such as psoriasis, PRP, or lymphoma. If the treatment of choice is anti-inflammatory, fast-acting drugs such as high-dose corticosteroids, infliximab, cyclosporine, or IL-17 inhibitors should be preferred.

4.1 Types of Erythroderma

4.1.1 Erythroderma in Children

Erythroderma in the newborn period is life-threatening. The most common cause is some form of lamellar ichthyosis or bullous ichthyosiform erythroderma. Many infants present as collodion babies and their disease evolves into erythroderma. A skin biopsy is usually the most rapid way to establish the diagnosis, if the family history does not provide the clue. Genetic tests are needed for confirmation. Leiner's disease is not a specific disease, but represents erythroderma in infants who are immunodeficient. Staphylococcal scalded skin syndrome is primarily a disease of children; the end stage is erythroderma, but the loss of sheets of skin generally leads to a diagnosis prior to total body involvement. In older children, psoriasis, atopic eczema, and PRP are the most common causes, just as in adults.

Bullous Erythroderma In addition to staphylococcal scalded skin syndrome other causes are bullous ichthyosis (today also known as epidermolytic ichthyosis) and rarely, generalized cutaneous mastocytosis.

Erythroderma with Scaling/Exfoliation This group is very heterogeneous. Primary immunodeficiency syndrome (Omenn syndrome and severe combined immunodeficiency) (▶ Chap. 62, "Hereditary Immunodeficiency Diseases: Skin Signs") and Comèl–Netherton syndrome should be considered, as well as lamellar ichthyosis, congenital ichthyosiform erythroderma, and harlequin ichthyosis (▶ Chap. 60, "Ichthyoses").

4.1.2 Drug-Induced Erythroderma

About 25% of erythroderma in adults is drug-induced (▶ Chap. 35, "Cutaneous Adverse Drug Reactions"). Although many different agents can cause erythroderma, some are notorious, including phenytoin, carbamazepine, captopril, cimetidine, diltiazem, nitrofurantoin, isoniazid, antimalarial agents, lithium salts, and gold salts. Drug reactions usually appear first as macular exanthems and over time coalesce into erythroderma. Patients tend to improve promptly when the suspected agent is withdrawn, although the erythroderma may still smolder for days to weeks. Sometimes, patients present more rapidly with widespread erythema. In either pattern, blisters may develop and the reaction may even evolve into toxic epidermal necrolysis (▶ Chap. 36, "Severe Cutaneous Adverse Drug Reactions").

Another type of drug-induced erythroderma is allergic contact dermatitis. When the reaction is severe enough, the process may spread outside the areas in which the offending agent was applied. This is referred to as hematogenous spread, which is not surprising because allergic contact dermatitis is mediated by circulating lymphocytes.

4.1.3 Erythroderma Associated with Hematological Malignancies

Today, there is a distinction between erythroderma in mycosis fungoides and Sézary's syndrome (▶ Chap. 110, "Cutaneous Lymphomas"). When patients with previously typical mycosis fungoides develop erythroderma it is a poor prognostic sign. In Sézary's syndrome the patient is erythrodermic, has circulating malignant T cells, and lymphadenopathy. Biopsies of the skin and lymph nodes usually help to establish the nature of the two diseases.

Erythroderma can also be a paraneoplastic marker for a wide range of hematological malignancies (Hodgkin's or non-Hodgkin's lymphomas and leukemia) and occasionally for other tumors. In such cases, the onset is more rapid and there is no history of precursor lesions. Histological examination of the lymph nodes or other tissue is crucial, because the skin changes are occasionally caused by infiltrating tumor cells.

4.1.4 Erythroderma Caused by Underlying Dermatoses

About 60% of erythroderma cases are caused by an exacerbation of an underlying dermatosis. In most instances, the characteristic signs of the skin disease are obscured by the intense inflammation, but often clues are to be found in the history or on examination of the oral mucosa or nails. Although there is a long list of diseases that have been reported to cause erythroderma, most cases arise in one of the disorders described in the following.

Atopic Eczema In younger patients, this is the most likely cause of erythroderma. Usually, a personal or family history of atopy is present. Pruritus is often extreme and associated secondary infections induced by scratching are common. Well-known atopic stigmata, such as Dennie–Morgan folds, palmar hyperlinearity, or nail changes, may help in the clinical diagnosis.

Psoriasis Although generalized pustular psoriasis is the usual precursor of psoriatic erythroderma, psoriasis vulgaris can also worsen and involve the entire body. Withdrawal of treatment with systemic steroids and occasionally cyclosporine can trigger the latter scenario, which develops more slowly than the pustular psoriasis outbreak.

Typical psoriatic nail changes and severe itching can become the main diagnostic symptoms.

Pityriasis Rubra Pilaris Even though this disease is uncommon in both children and adults, some reports indicate that it may account for a disproportionate share of the erythroderma cases (approximately 1%). The presence of areas that have been spared with keratotic papules at their periphery and of thickened palms and soles are two good clinical clues.

Seborrheic Dermatitis When the elderly and infirm are either unable to care for their scalp and skin or are in a nursing home where such care is not adequate, their previously easily managed seborrheic dermatitis may flare dramatically. In this population, seborrheic dermatitis is one of the most common precursors of erythroderma, although one should always search for an underlying malignancy and evaluate the patient's medications carefully, as often the history of seborrheic dermatitis distracts from the correct diagnosis.

Lichen Planus The nails and oral mucosa are the best places to search for clues to lichen planus. Erythroderma is a very rare complication.

Pemphigus Foliaceus The face and upper part of the chest are most often involved initially. Although pemphigus foliaceus is a blistering disorder, it may present primarily with erythroderma. Clues include moist scaling, occasional circular areas of scaling that represent attempts at blister formation, and generally a long history of some sort of facial rash.

Crusted Scabies Infections with *Sarcoptes scabiei* may lead to very widespread disease particularly in elderly, immunodeficient patients, or patients with special needs (e.g., Down's syndrome). Those with HIV/AIDS are also at risk. Not surprisingly, the pruritus is extreme. Although erythroderma is described, there is usually massive crusting, especially on the hands and feet. Often there is an eosinophilia. Nursing personnel or contacts also frequently have itching. Inappropriate care may worsen the erythroderma. *S. scabiei* are present in huge numbers and can easily be identified.

Idiopathic Erythroderma (Red Man Syndrome) Describing erythroderma as idiopathic generally means that one has failed to identify the underlying disorder. A possible exception is red man syndrome. The name causes confusion because the French *l'homme rouge* is also used for Sézary's syndrome. The original name red man syndrome goes back to Thestrup-Pedersen (Thestrup-Pedersen et al. 1988), who described various causes with the same clinical picture for the first time.

Patients are almost exclusively older men who have intense pruritus, hyperkeratotic palms and soles (79% of cases), and elevated IgE levels (69% of cases; Fig. 18). Lymphadenopathy is common and weight loss may occur.



Fig. 18 Idiopathic erythroderma

Although suggestive of mycosis fungoides or Sézary's syndrome, in many cases, peripheral blood and histological evaluation fail to confirm the suspicion. Some patients even have a monoclonal T cell proliferation, but fail to advance to overt disease. Careful clinical control and follow-up laboratory evaluation are mandatory, as transition to malignant disease has occurred, even after many years.

Melanoerythroderma This rare form of erythroderma mainly affects men. A reddening and pigmentation of the entire skin can be observed. After the redness subsides, melanoderma develops (often with islands of depigmented skin). In most cases there is a swelling of the lymph nodes and cachexia (Gartmann–Tritsch syndrome). Hair loss and onychodystrophy occur.

Other Associations

Erythroderma has also been described in rare cases of lupus erythematosus, dermatophytosis, hepatitis, renal failure, AIDS, Omenn syndrome, graft-versus-host disease, histoplasmosis, thyrotoxicosis, sarcoidosis, and as vancomycin-associated erythroderma.

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